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TECHNOLOGICAL UNIVERSITY DUBLIN

CITY CAMPUS - GRANGEGORMAN

Master of Science in Computer Science –
(Advanced Software Development - Full-Time)

Master of Science in Computer Science –
(Data Science - Part-Time)

Year 1

SEMESTER 1 - EXAMINATIONS 2024/25

Data Mining

Internal Examiner:
Dr Luis Miralles
Dr Paul Doyle

External Examiner:
Mr Alan McClean
Professor Michaela Black

Exam Duration: 2 hours

Instructions: All questions are compulsory

Special Instructions/Handouts: No calculators or other devices should be used.

- Q1. (a) Describe the steps you would follow to apply K-means clustering to customer segmentation in an online retail website business. (10 marks)

1. **Data Collection and Preparation:** Gather and clean customer data (e.g., purchase history, demographics) and normalize features for consistent scaling.
2. **Determine Number of Clusters (K):** Use methods like the elbow method or silhouette score to find the optimal K.
3. **Initialize and Assign Clusters:** Randomly initialize K centroids, then assign each data point to its nearest centroid.
4. **Update and Iterate:** Recalculate centroids by averaging assigned points, and repeat until assignments stabilize.
5. **Interpret and Validate Clusters:** Analyse cluster insights for actionable strategies and validate cluster quality using metrics like silhouette score.

- (b) Explain the importance of data normalization and data integration in machine learning. (10 marks)

- **Data Normalization:** Ensures all features contribute equally, improving model performance and convergence speed in algorithms sensitive to feature scale (e.g., K-means, neural networks).
- **Data Integration:** Combines data from multiple sources, enhancing data quality and providing a comprehensive dataset, which leads to more accurate model insights.

- (c) Differentiate between feature selection and feature extraction. Provide an example of each in a classification problem. (10 marks)

- **Feature Selection:** Identifying and selecting the most relevant features from the original data, reducing dimensionality without creating new features (e.g., selecting top features based on correlation).
- **Feature Extraction:** Creating new features by transforming or combining existing ones (e.g., using Principal Component Analysis (PCA) to generate a lower-dimensional set of features that captures maximum variance).receiving an update.

- Q2 (a) Describe the following time series similarity metrics Dynamic Time Warping (DTW) and Symbolic Aggregate Approximation (SAX). (10 marks)

- **Dynamic Time Warping (DTW):** DTW aligns two time series by stretching or compressing segments to minimize the distance between them, allowing comparison despite variations in speed or length. It's widely used in tasks like speech recognition, where similar patterns may occur at different speeds.
- **Symbolic Aggregate Approximation (SAX):** SAX reduces the dimensionality of a time series by converting it into a string of symbols, representing trends over time while preserving shape and reducing noise. This technique is beneficial for indexing, clustering, and searching large time series datasets.

- (b) Discuss how LIME and SHAP address model explainability and how they differ in their approach.

(10 marks)

- **LIME (Local Interpretable Model-agnostic Explanations):** Generates locally faithful explanations by creating a simpler model around the prediction of interest, providing instance-level interpretations.
- **SHAP (SHapley Additive exPlanations):** Provides global and local interpretability by calculating the contribution of each feature to predictions based on game-theoretic Shapley values.

- (c) Indicate if each statement is true or false and provide an explanation for your choice:

1. Supervised Learning requires labeled data for training.
2. Feature extractions techniques like Recursive Feature Elimination reduce the dimensionality by transforming features rather than selecting subsets of original features.
3. Decision Trees can be used for both classification and regression tasks, providing flexibility in supervised learning applications.
4. SHAP and LIME are model-specific explainability techniques.

(10 marks)

1. **True**, as supervised learning relies on labelled input-output pairs.
2. **False**, RFE is a selection technique.
3. **False**, time series can exhibit both linear and non-linear patterns.
4. **False**, both are model-agnostic and can be applied to various model types.

Q3 Answer four out of the five next points:

- (a) Explain how the Apriori Algorithm uses Support, Confidence, and Lift to generate meaningful association rules, and provide a practical example of its use.

(10 marks)

1. **Support:** The frequency of itemsets in the dataset (e.g., the percentage of transactions containing bread).
2. **Confidence:** The likelihood of an item being bought if another item is bought (e.g., if a customer buys bread, they are likely to buy butter).
3. **Lift:** The ratio of observed support to that expected if the items were independent. Higher lift values indicate stronger associations.

- (b) Describe the difference between TF-IDF and word embeddings, and discuss in what type of NLP application each might be more suitable. (10 marks)

- **TF-IDF:** Captures the importance of words in a document by emphasizing rare terms. Suitable for document retrieval and classification tasks.
- **Word Embeddings:** Encodes words in a dense, continuous vector space, capturing semantic relationships. Best for sentiment analysis, translation, and semantic similarity tasks.

- (c) Indicate if each statement is true or false and provide an explanation for their choice:
- Data Mining and Machine Learning are synonymous terms that refer to the same concepts.
 - K-means clustering requires prior knowledge of the number of clusters to be formed.
 - Data preprocessing is an optional step in the Data Mining life cycle.
 - The CRISP-DM framework is the only methodology used for Data Mining projects.

• (10 marks)

- **False,** Data Mining focuses on discovering patterns in data, while Machine Learning is about creating predictive models.
- **True,** the number of clusters (K) is specified before running K-means.
- **False,** it is essential to ensure data quality and improve analysis accuracy.
- **False,** other frameworks exist, like SEMMA and KDD.

- (d) Compare and contrast supervised and unsupervised learning. Provide examples of algorithms used in each category and discuss their typical applications. (10 marks)

Supervised Learning:

- Definition: Learning from labeled data where the model is trained to map inputs to known outputs.
- Examples of Algorithms:
 - Linear Regression (for predicting continuous values).
 - Decision Trees (for classification tasks).
- Typical Applications: Spam detection, credit scoring, and medical diagnosis.

Unsupervised Learning:

- Definition: Learning from unlabeled data, aiming to discover hidden patterns or groupings.
- Examples of Algorithms:
 - K-means Clustering (for grouping data).
 - Principal Component Analysis (for dimensionality reduction).
- Typical Applications: Customer segmentation, anomaly detection, and topic modeling.

- (e) Explain why Logistic Regression is preferred over Linear Regression for binary classification, including an example of each model's use. (10 marks)

- **Logistic Regression:** Maps output to probabilities, making it ideal for binary classification. E.g., predicting if an email is spam or not.
- **Linear Regression:** Predicts continuous values, such as predicting house prices based on square footage.